

School Management System

Complete Product Requirements Document (PRD)

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📋 Table of Contents

1. Executive Summary
2. Product Overview
3. User Roles & Features
4. Technical Architecture
5. Database Design
6. Security & Compliance
7. Multi-Tenancy & White-Labeling
8. Monetization Strategy
9. Development Roadmap
10. Resources Required
11. Go-To-Market Strategy
12. Appendix

1. Executive Summary

1.1 Vision Statement

To create a comprehensive, secure, and user-friendly School Management System that digitizes school operations, improves communication between stakeholders, and provides actionable insights for better educational outcomes.

1.2 Problem Statement

Current Challenges	Impact
Manual attendance tracking	Time-consuming, error-prone
Paper-based record keeping	Risk of data loss, hard to search
Poor parent-school communication	Parents uninformed, increased calls
Manual fee collection	Cash handling risks, tracking issues
No real-time visibility	Delayed decision making
Report card generation	Hours of manual work

1.3 Solution

A unified digital platform connecting **Administrators**, **Teachers**, **Students**, and **Parents** through web and mobile applications, automating routine tasks and enabling real-time communication.

1.4 Key Benefits

Stakeholder	Benefits
School Admin	Centralized control, automated reports, fee tracking

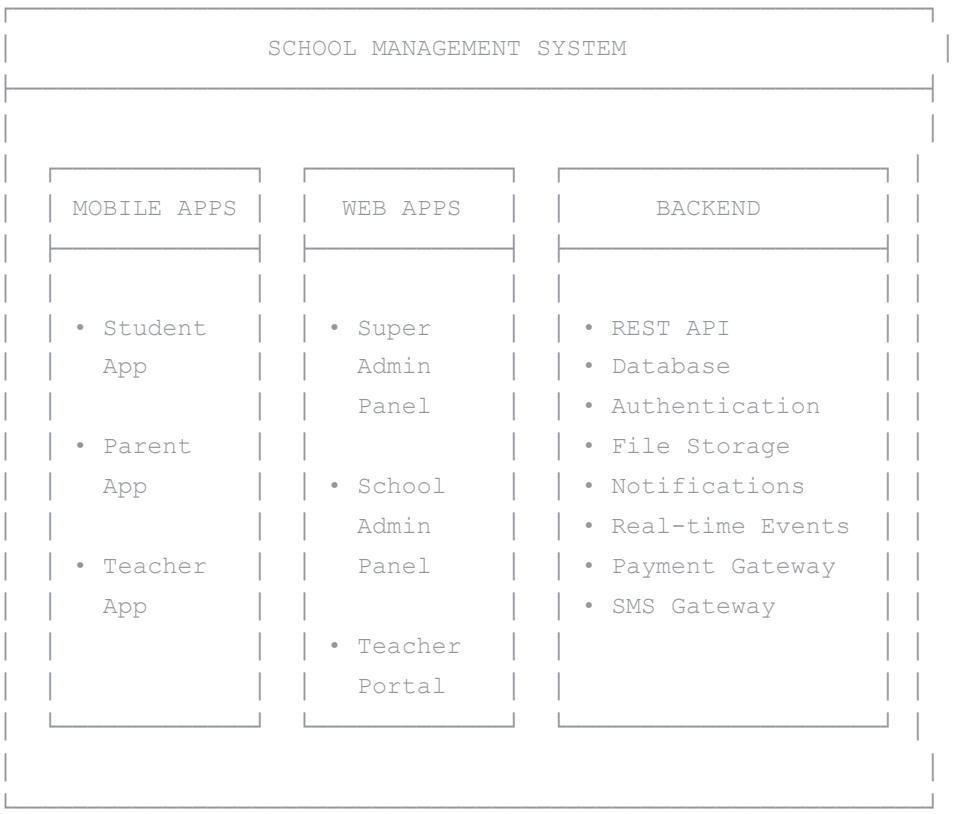
Teachers	Quick attendance, easy grade entry, parent communication
Students	Access to materials, homework, results anytime
Parents	Real-time updates, fee payment, direct teacher contact

1.5 Market Opportunity

- Global EdTech market: **$404 billion by 2025**
- Less than **3% of schools** currently use digital management systems
- Growing demand for parent engagement tools
- COVID-19 accelerated digital adoption in education

2. Product Overview

2.1 Product Components



2.2 Platform Support

Platform	Technology	Users
Android	Flutter	Students, Parents, Teachers
iOS	Flutter	Students, Parents, Teachers
Web (Admin)	React / Next.js	Super Admin, School Admin
Web (Teacher)	React / Next.js	Teachers (optional)

2.3 Core Modules

Module	Description	Priority
User Management	Students, teachers, parents, staff	🔴 High
Attendance	Daily attendance marking & reports	🔴 High
Academics	Classes, subjects, timetable	🔴 High
Examinations	Exams, results, report cards	🔴 High
Fee Management	Fee structure, collection, receipts	🔴 High
Communication	Announcements, notifications, chat	🔴 High
Homework	Assignment creation & submission	🟡 Medium
Transport	Routes, tracking, alerts	🟡 Medium
Library	Book inventory, issue/return	🟢 Low
Inventory	Assets, maintenance	🟢 Low

3. User Roles & Features

3.1 Super Admin (Platform Owner)

Access: Web Panel Only

Module	Features
Dashboard	Total schools, students, revenue, growth charts
School Management	Add/edit/deactivate schools
Subscription	Plans, billing, invoices, payment tracking
User Management	Support staff accounts
System Settings	Global configurations, feature toggles
Support	View/respond to school tickets
Reports	Revenue reports, usage analytics

3.2 School Admin

Access: Web Panel + Mobile App

3.2.1 Dashboard

- Quick statistics (students, teachers, attendance today)
- Fee collection summary
- Recent activities
- Pending tasks & alerts
- Calendar with upcoming events

3.2.2 Student Management

Feature	Description

Add Student	Individual or bulk import (Excel)
Student Profile	Personal details, photo, documents
Admission	Admission number generation
Promotion	Bulk promotion to next class
Transfer	Transfer certificate generation
ID Cards	Generate & print student ID cards

3.2.3 Fee Management

Feature	Description
Fee Structure	Define fee types & amounts per class
Fee Collection	Record payments, generate receipts
Online Payment	Integrate payment gateway
Due Reminders	Automated SMS/push reminders
Reports	Collection reports, defaulter lists
Discounts	Sibling discount, scholarship management

3.3 Teacher

Access: Mobile App + Web Portal (Optional)

Module	Features
Dashboard	Today's classes, pending tasks, announcements
Attendance	Mark attendance (list/grid view), bulk marking
Academics	View timetable, assigned classes, substitutions
Homework	Create homework, view submissions, grade

Examinations	Enter marks, view class performance
Communication	Message parents, create announcements
Leave Management	Apply leave, view status, leave balance

3.4 Student

Access: Mobile App Only

Module	Features
Dashboard	Today's timetable, pending homework, announcements
Attendance	View daily attendance, monthly summary, percentage
Timetable	Weekly class schedule
Homework	View homework, submit completed work
Exams	View date sheet, results, report card
Fee	View fee details, payment history, download receipts
Study Materials	Access notes, videos, resources
Transport	Bus route, timing, driver contact

3.5 Parent

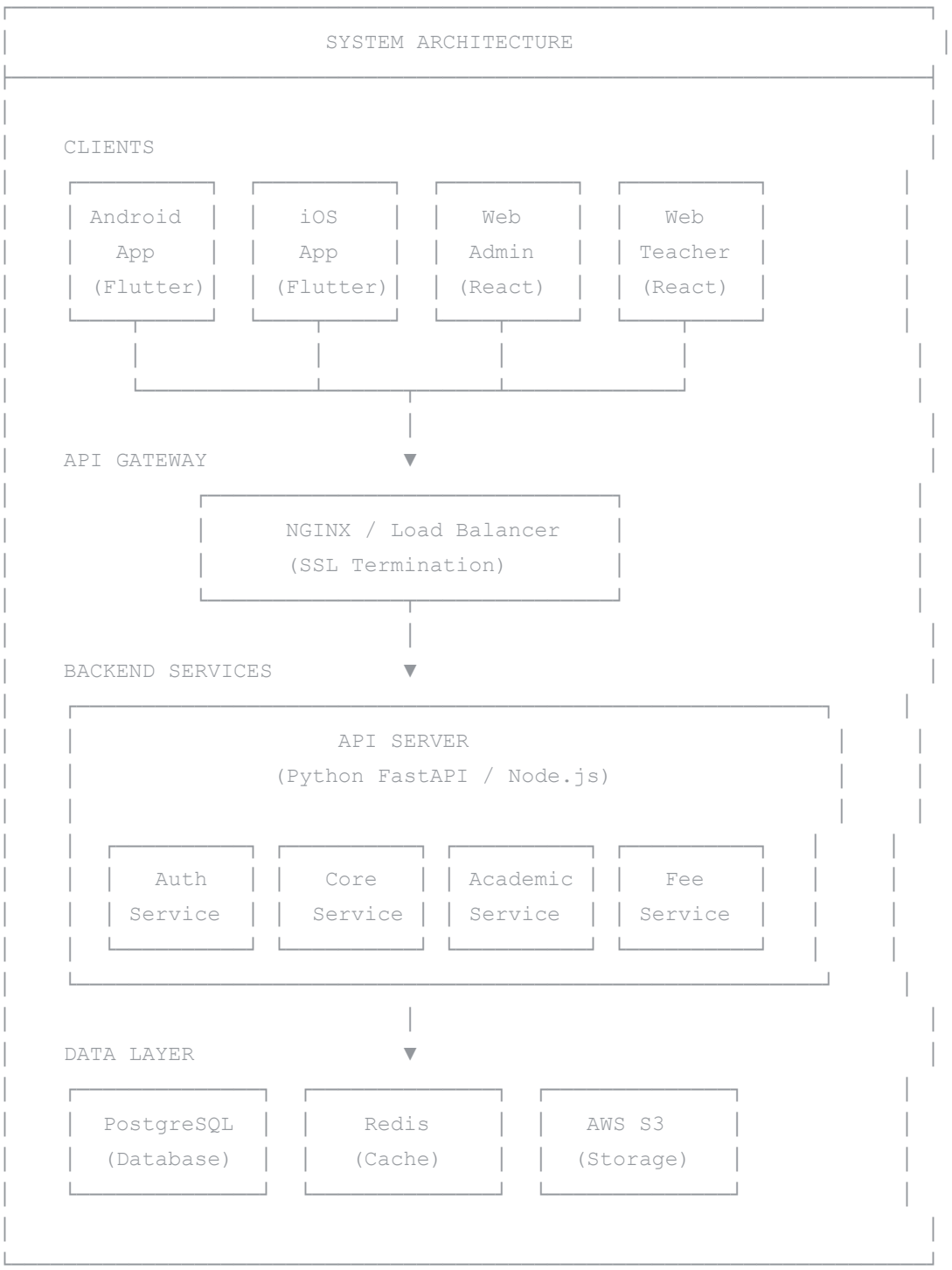
Access: Mobile App Only

Module	Features
Dashboard	Child's summary, fee due alerts, notifications
Children	Switch between multiple children
Attendance	Daily status, monthly report, alerts for absence
Academics	View grades, report cards, class rank

Fee	View dues, pay online, download receipts
Communication	Chat with class teacher, view announcements
Transport	Live bus tracking, arrival alerts
Leave	Apply for child's leave

4. Technical Architecture

4.1 System Architecture



4.2 Technology Stack

Mobile Applications

Component	Technology	Justification
Framework	Flutter 3.x	Single codebase for Android & iOS
State Management	Riverpod / Bloc	Scalable, testable
Local Database	Hive / SQLite	Offline data support
HTTP Client	Dio	Robust API handling
Push Notifications	Firebase FCM	Free, reliable

Backend

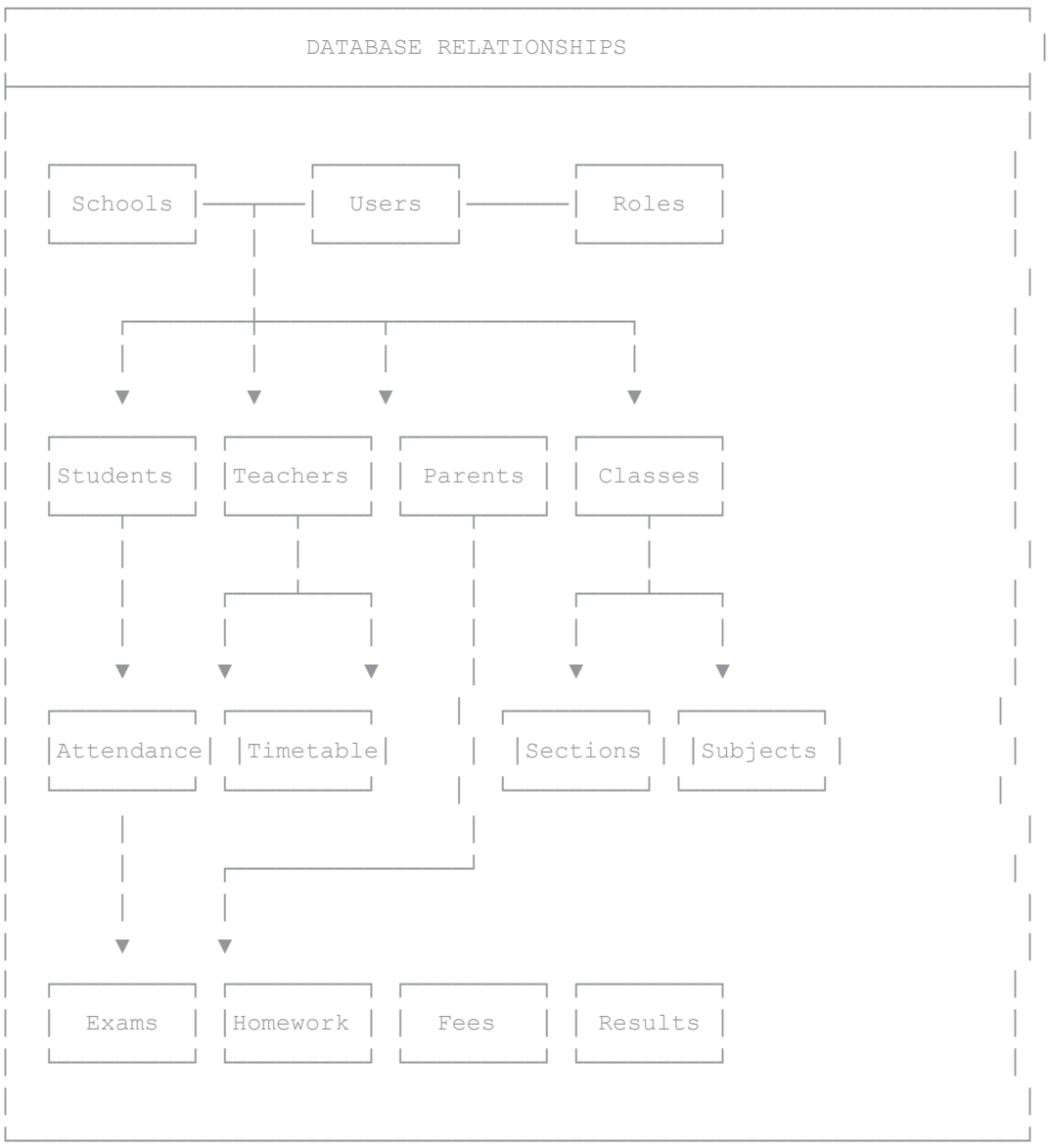
Component	Technology	Justification
Language	Python 3.11+	Rapid development
Framework	FastAPI	High performance, auto docs
ORM	SQLAlchemy	Database abstraction
Authentication	JWT + OAuth2	Secure, stateless
Task Queue	Celery + Redis	Background jobs

Database & Storage

Component	Technology	Justification
Primary DB	PostgreSQL 15	Robust, relational
Cache	Redis	Fast caching, sessions
File Storage	AWS S3 / Cloudflare R2	Scalable, cheap

5. Database Design

5.1 Entity Relationship Overview



5.2 Core Tables

Schools Table

```
CREATE TABLE schools ( id SERIAL PRIMARY KEY, code VARCHAR(50) UNIQUE NOT NULL,
name VARCHAR(255) NOT NULL, address TEXT, city VARCHAR(100), phone VARCHAR(20),
email VARCHAR(255), logo_url VARCHAR(500), primary_color VARCHAR(7) DEFAULT
```

```
'#1976D2', plan VARCHAR(50) DEFAULT 'trial', plan_expires_at TIMESTAMP,
max_students INTEGER DEFAULT 50, is_active BOOLEAN DEFAULT true, created_at
TIMESTAMP DEFAULT CURRENT_TIMESTAMP );
```

Students Table

```
CREATE TABLE students ( id SERIAL PRIMARY KEY, school_id INTEGER REFERENCES
schools(id) NOT NULL, admission_number VARCHAR(50) NOT NULL, first_name
VARCHAR(100) NOT NULL, last_name VARCHAR(100), date_of_birth DATE, gender
VARCHAR(10), class_id INTEGER REFERENCES classes(id), section_id INTEGER
REFERENCES sections(id), parent_id INTEGER REFERENCES parents(id), status
VARCHAR(20) DEFAULT 'active', created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
UNIQUE(school_id, admission_number) ); CREATE INDEX idx_students_school ON
students(school_id);
```

Attendance Table

```
CREATE TABLE attendance ( id SERIAL PRIMARY KEY, school_id INTEGER REFERENCES
schools(id) NOT NULL, student_id INTEGER REFERENCES students(id) NOT NULL, date
DATE NOT NULL, status VARCHAR(20) NOT NULL, marked_by INTEGER REFERENCES
teachers(id), marked_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP, UNIQUE(student_id,
date) ); -- Status: 'present', 'absent', 'late', 'half_day'
```

6. Security & Compliance

6.1 Security Architecture



6.2 Data Isolation (Multi-Tenancy)

Key Principle: School A can NEVER see School B's data.

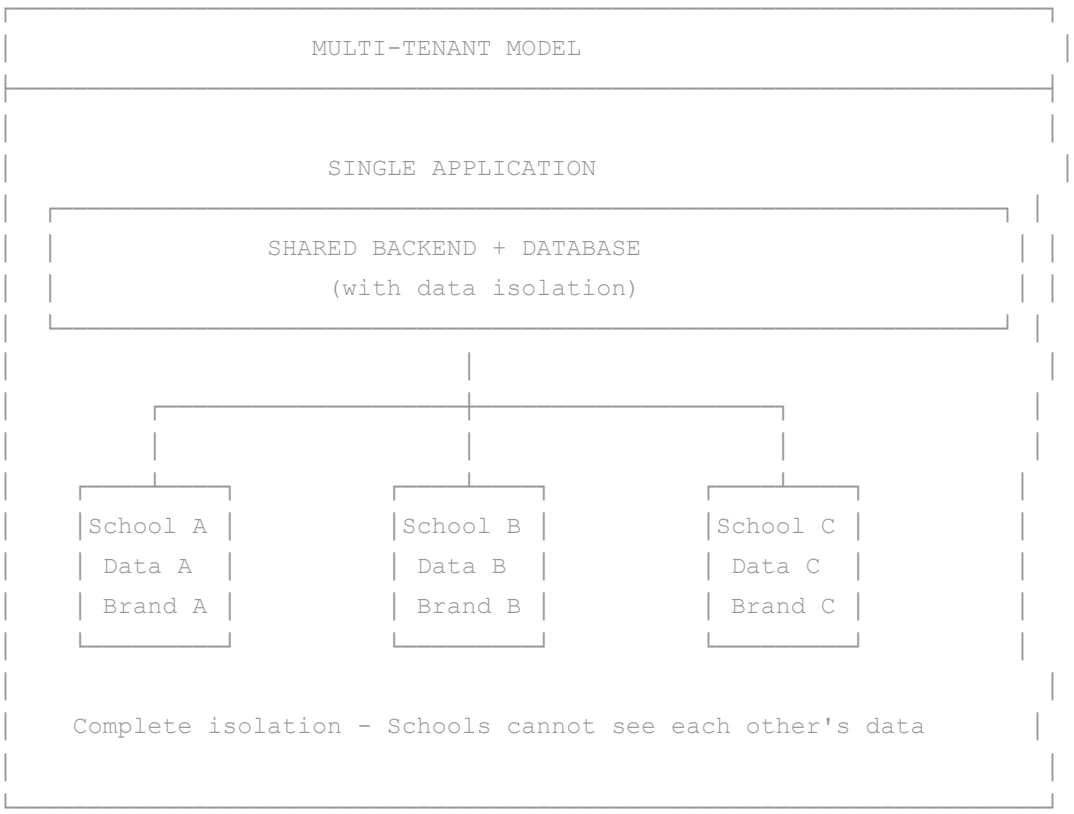
- Every table has `school_id` column
- Every API request validates user's school access
- Row-Level Security enforced at database level
- Complete audit trail of all data access

6.3 Trust-Building Features

Feature	Purpose
Data Export	Schools can export all their data anytime
Audit Logs	Complete history of who did what
Backup Policy	Every 6 hours, 30-day retention
Privacy Policy	Clear data handling documentation
No Data Selling	Revenue from subscriptions only

7. Multi-Tenancy & White-Labeling

7.1 Multi-Tenant Architecture



7.2 White-Label Customization

Element	Customizable	Example
School Logo	✅ Yes	School's own logo
App Name	✅ Yes	"ABC High School App"
Primary Color	✅ Yes	School's brand color
Login Screen	✅ Yes	Custom background
Email Templates	✅ Yes	School's header/footer

SMS Sender ID	✅ Yes	"ABCHIGH"

7.3 School Onboarding Process

Step	Action	Owner
1	Contract signed, payment received	Sales
2	Create school in super admin panel	You
3	Configure branding (logo, colors)	You
4	Create admin account, send credentials	System
5	School admin sets up classes, imports data	School
6	Training session (video call)	You
7	Go live	School

8. Monetization Strategy

8.1 Pricing Tiers

STARTER

$99

/month

Up to 200 students

- ✓ Core modules
- ✓ Mobile apps
- ✓ Email support
- ✓ 5GB storage

PRO

$249

/month

Up to 1000 students

- ✓ All Starter features
- ✓ Online classes
- ✓ Transport module
- ✓ Priority support
- ✓ 25GB storage

ENTERPRISE

$499+

/month

Unlimited students

- ✓ All Pro features
- ✓ White-label
- ✓ Custom domain
- ✓ API access
- ✓ Dedicated support

8.2 Additional Revenue Streams

Revenue Source	Model	Example
SMS Charges	Markup on usage	$0.02/SMS (cost: $0.01)
Payment Gateway	Commission	0.5% of fee collected
Setup Fee	One-time	$200-500
Training	Per session	$100-300
Custom Features	Fixed/Hourly	Based on scope

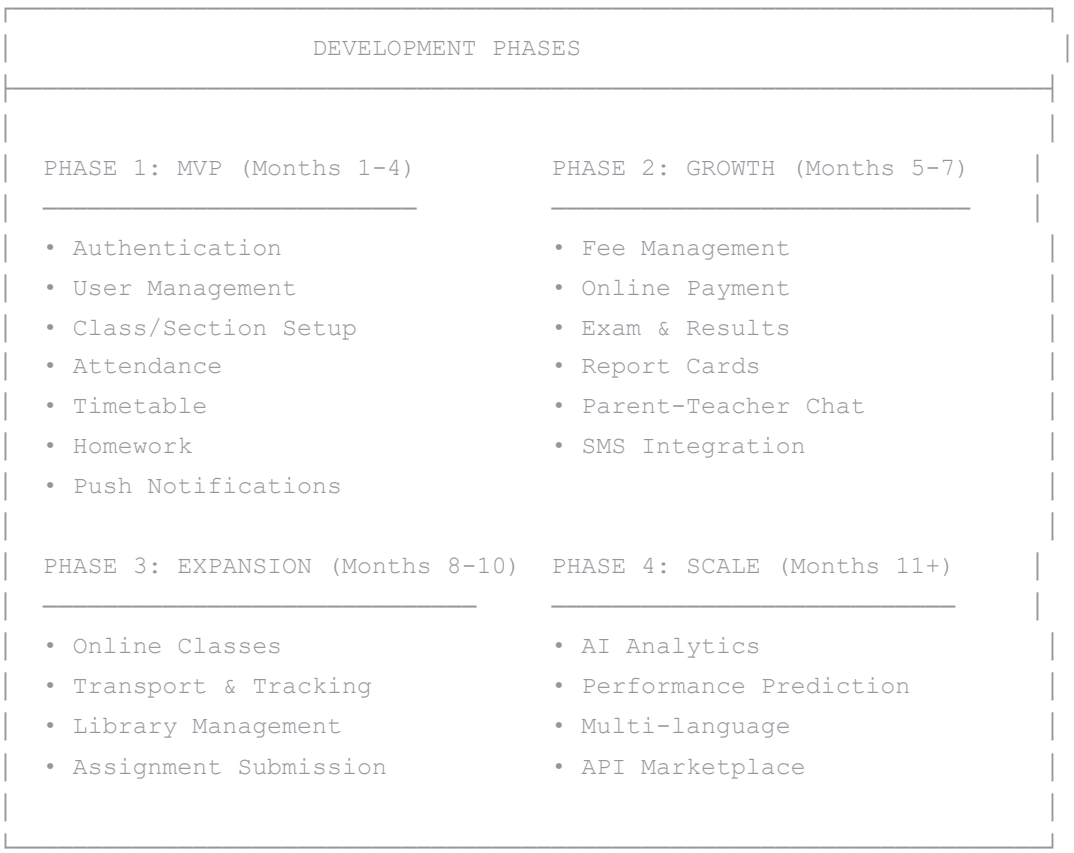
8.3 Revenue Projections

Timeline	Schools	Avg Revenue	Monthly Revenue	Annual Revenue

Month 6	5	$150	$750	$9,000
Year 1	25	$180	$4,500	$54,000
Year 2	100	$200	$20,000	$240,000
Year 3	300	$220	$66,000	$792,000

9. Development Roadmap

9.1 Phase Overview



9.2 Detailed Timeline - Phase 1 (MVP)

Month	Week	Tasks
Month 1	1	Project setup, database design, API structure
	2	Authentication system (login, register, JWT)
	3	School & user management APIs
	4	Class, section, subject APIs
Month 2	1	Admin panel setup, authentication UI

	2	Dashboard, school settings
	3	Student management UI
	4	Teacher management UI
Month 3	1	Flutter project setup, authentication
	2	Student app - dashboard, attendance, timetable
	3	Parent app - dashboard, child info, notifications
	4	Teacher app - attendance marking, homework
Month 4	1	Push notifications, announcements
	2	Integration testing
	3	Bug fixes, performance optimization
	4	Pilot launch with 2-3 schools

10. Resources Required

10.1 Team Structure (Full Team)

Role	Count	Monthly Cost	Responsibility
Project Manager	1	$1,000-2,000	Planning, coordination
Backend Developer	1-2	$1,500-3,000	APIs, database
Flutter Developer	1-2	$1,500-3,000	Mobile apps
Frontend Developer	1	$1,200-2,500	Admin panel
UI/UX Designer	1 (part-time)	$500-1,000	Design
QA Tester	1 (part-time)	$500-800	Testing
Total		**$6,200 - $12,300/month**	

10.2 Infrastructure Costs (Monthly)

Service	Provider	Starter	Growth
Server	DigitalOcean	$20	$100
Database	Managed PostgreSQL	$15	$50
Cache	Managed Redis	$10	$30
File Storage	AWS S3	$5	$30
CDN	Cloudflare	Free	$20
Email	SendGrid	Free	$20
Push Notifications	Firebase	Free	Free

Total		**~$50/month**	**~$250/month**

10.3 Third-Party Services

Service	Provider	Cost
SMS Gateway	MSG91 / Twilio	$0.01-0.03 per SMS
Payment Gateway	Razorpay / Stripe	2% per transaction
Video Calls	Agora / 100ms	Free tier, then usage-based
Maps (Bus Tracking)	Google Maps	$200 free credit/month
Error Tracking	Sentry	Free tier

11. Go-To-Market Strategy

11.1 Launch Phases

Phase 1: Pilot (Month 1-2 after development)

- Offer **free to 3-5 schools** for 3 months
- Get real-world feedback
- Fix bugs and improve UX
- Collect testimonials and case studies

Phase 2: Local Launch (Month 3-6)

- Target your city/region first
- Personal school visits
- Demos and free trials
- Partner with school associations

Phase 3: Scale (Month 6+)

- Digital marketing (Google Ads, Facebook)
- Content marketing (blog, YouTube)
- Referral program
- Reseller partnerships

11.2 Sales Approach

Target Decision Makers

- School Principal
- School Owner / Trustee
- Administrative Head

Sales Pitch - Address Pain Points

Pain Point	Your Solution

"Parents keep calling for updates"	"Parents get real-time notifications"
"Attendance takes too long"	"One-tap attendance in 2 minutes"
"Report cards take weeks"	"Auto-generated in seconds"
"Fee tracking is messy"	"Complete fee management with reminders"
"No visibility into operations"	"Real-time dashboard with all metrics"

11.3 Convincing Schools - Security Answers

Question	Your Answer
"Is our data safe?"	"Encrypted, backed up every 6 hours, stored on secure cloud servers - safer than any paper system"
"Can other schools see our data?"	"Complete isolation. Each school is like a separate private system"
"What if you shut down?"	"Export your data anytime. No lock-in. Your data is always yours"
"Why should we trust you?"	"14-day free trial. See it working before paying anything"

12. Appendix

12.1 Competitor Comparison

Feature	Your App	Fedena	Teachmint	ClassDojo
Mobile Apps	✅	Limited	✅	✅
Fee Management	✅	✅	✅	❌
Online Classes	✅	❌	✅	❌
Bus Tracking	✅	❌	❌	❌
White-Label	✅	❌	❌	❌
Affordable	✅	❌	✅	✅
Modern UI	✅	❌	✅	✅

12.2 Success Metrics

Metric	Target (Year 1)
Schools Onboarded	25
Monthly Recurring Revenue	$4,500
Customer Retention	90%
NPS Score	50+
App Store Rating	4.5+

12.3 Glossary

Term	Definition

Multi-Tenancy	Single system serving multiple organizations
White-Labeling	Customizing branding per customer
SaaS	Software as a Service (subscription model)
JWT	JSON Web Token (authentication)
RBAC	Role-Based Access Control
API	Application Programming Interface
FCM	Firebase Cloud Messaging

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